LESSON 9.8

DILATIONS ON THE COORDINATE PLANE

LESSON INTRODUCTION

MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

LEARNING TARGETS

- I can describe how an image and preimage generated by a single dilation are related.
- I can apply the effect of a dilation on the coordinate plane.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.2 Identify transformations that do or do not preserve distance

ESSENTIAL QUESTIONS

- How does the center of dilation impact the coordinates of the image?
- How does the scale factor impact the coordinates of the image?
- How are the image and preimage of a dilation related?

LESSON NARRATIVE

Students focus on dilations on a coordinate plane. Students discover that when the center of dilation is the origin, the coordinates of the preimage are multiplied by the scale factor to determine the coordinates of the image. They also discover that the preimage and image are similar figures.

COMPONENT SUMMARY

9.8.1 WARM-UP *(~5 MIN)*

Differentiate between triangles

9.8.2 EXPLORATION *(15 MIN)*

Explore dilations on coordinate plane

9.8.3 BRING IT TOGETHER *(10 MIN)*

Consolidate understanding of dilations on coordinate plane

9.8.4 GUIDED PRACTICE *(5 MIN)*

Describe relationship between preimage and image

9.8.5 YOUR TURN! *(5 MIN)*

Practice performing a dilation on coordinate plane

9.8.6 WRAP-UP *(~5 MIN)*

Reflect on mastery of learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Dilation
- Scale Factor
- Center of Dilation
- Similar Figures

MATERIALS

- (Optional) Patty Paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly think that the angles are multiplied by the scale factor.
- Students may incorrectly think that a dilation always results in a larger image.

THINGS TO CONSIDER

- Consider prompting students to measure the angles to discover that they are congruent.
- If time is an issue, then consider assigning Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong

In 9.8.1 Warm-Up students are presented with four triangles on a grid with the prompt "Which one doesn't belong?" Each of the four options "doesn't belong" for a different reason, and the similarities and differences are mathematically significant. Ask students to explain their rationales and provide opportunities to make their rationales more precise.

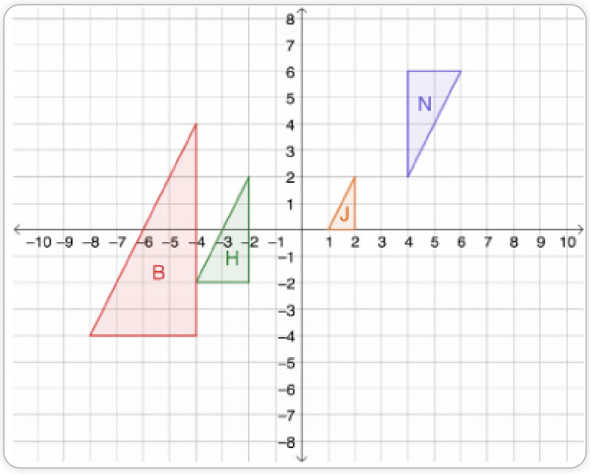
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 9.8.2 Exploration students explore the patterns between the coordinates of the preimage and image. Students recognize the structure that the coordinates of the preimage are multiplied by the scale factor. They also discover that scale factors greater than one result in a larger image and scale factors less than one result in smaller images.

DIFFERENTIATE INSTRUCTION

The discussions during 9.8.2 Exploration provide an auditory entry-point. 9.8.4 Guided Practice provides a visual entry-point as students see several examples of dilations. Provide a kinesthetic entry-point by prompting students to sketch the image on patty paper and then align the corresponding angles one at a time to the preimage to allow students to see that the angles fit perfectly one on top of the other in 9.8.3 Bring It Together. Finally, students can cut out the triangles in 9.8.4 Guided Practice to compare the angles and side lengths.



9.8.1 WARM-UP: WHICH ONE DOESN'T BELONG?

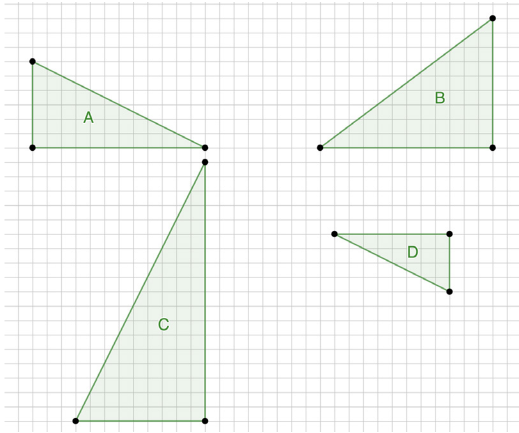
TEACHER MOVES

Which One Doesn't Belong

Students are presented with four triangles on a grid with the prompt "Which one doesn't belong?" Each of the four options "doesn't belong" for a different reason, and the similarities and differences are mathematically significant. Ask students to explain their rationales and provide opportunities to make their rationales more precise.

9.8.1 Warm-Up: Which One Doesn't Belong?

1. Four triangles are shown.

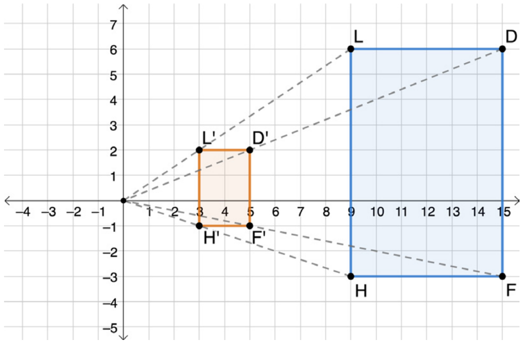


   a. Which one doesn't belong?
   Answers will vary.

   b. Explain your reasoning.
   Answers will vary. Triangle B doesn't belong because for all other triangles, the ratios of the short sides to the medium sides is 0.5*, but for triangle B, the ratio of the short side to the medium side is* 0.75*.*

9.8.2 Exploration: Dilations on the Coordinate Plane

1. On the coordinate plane, quadrilateral $DLHF$ was dilated with a scale factor of $\frac{1}{3}$ centered at the origin.



a. Complete the table.

Sides of the Preimage	Length	Sides of the Image	Length
Segment LD	6	Segment $L'D'$	2
Segment DF	9	Segment $D'F'$	3
Segment HF	6	Segment $H'F'$	2
Segment LH	9	Segment $L'H'$	3

b. Complete the statement.

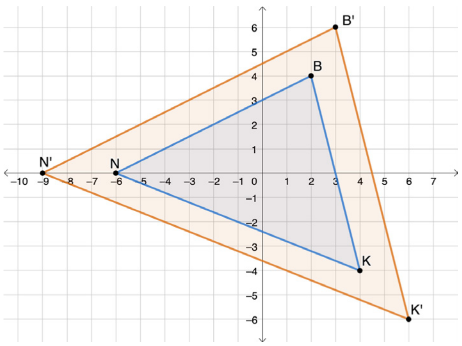
The side lengths of quadrilateral $D'L'H'F'$ are ○ $\frac{1}{3}$ ○ 3 times the length of the sides of quadrilateral $DLHF$.

c. Complete the table with the coordinates of the vertices of the preimage and image.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point D	$(15, 6)$	Point D'	$(5, 2)$
Point L	$(9, 6)$	Point L'	$(3, 2)$
Point H	$(9, -3)$	Point H'	$(3, -1)$
Point F	$(15, -3)$	Point F'	$(5, -1)$

d. What do you notice about the relationship between the coordinates of the corresponding points on the preimage and image?
Answers will vary. The coordinates of the image are $\frac{1}{3}$ times the coordinates of the corresponding points on the preimage.

2. On the coordinate plane, $\Delta B'K'N'$ is the result of a dilation of ΔBKN centered at the origin. The scale factor is 1.5.



9.8.2 EXPLORATION: DILATIONS ON THE COORDINATE PLANE

TEACHER MOVES

This exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There are opportunities throughout the Exploration for the student to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group Bring it Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

ENRICHMENT

- How did the distance from the origin of the corresponding points change?
- How did the side lengths change? How did the angles change?

DIFFERENTIATE INSTRUCTION

The discussions during this exploration provide an auditory entry-point. Prompt students to discuss how they obtained their answers with their partners. Ask students to consider if they agree with their partners' justifications and why or why not.

9.8.2 EXPLORATION: DILATIONS ON THE COORDINATE PLANE (CONTINUED)

TEACHER MOVES

Patterns and Structures

Students explore the patterns between the coordinates of the preimage and image. Students recognize the structure that the coordinates of the preimage are multiplied by the scale factor. They also discover that scale factors greater than one result in a larger image and scale factors less than one result in smaller images.

ENRICHMENT

When would a dilation yield an image congruent to the preimage?

a. Complete the table with the coordinates of the vertices of the preimage and image.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point B	$(2, 4)$	Point B'	$(3, 6)$
Point K	$(4, -4)$	Point K'	$(6, -6)$
Point N	$(-6, 0)$	Point N'	$(-9, 0)$

b. Discuss with your partner how the relationship between the coordinates is related to the scale factor. Summarize your discussion.
Answers will vary. The ratio between the coordinates of the image and the corresponding coordinates of the preimage is equal to the scale factor.

3. Suppose point W is the vertex of a polygon and has coordinates $(2, 10)$.

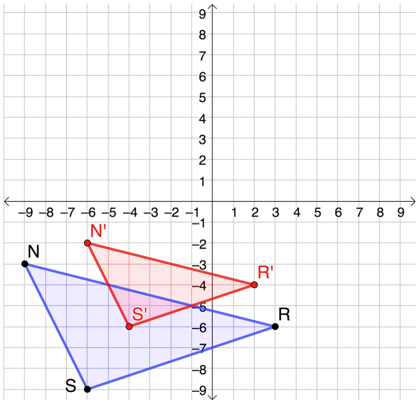
   a. If this polygon were dilated by a scale factor of 2.7 centered at the origin, discuss with your partner how you could determine the coordinates of the image of point W.

   b. Determine the coordinates of point W'.
   $(5.4, 27)$

9.8.3 Bring it Together: Dilations on the Coordinate Plane

When a dilation is performed on a figure on the coordinate plane with its center at the origin, the coordinates of the image are equal to the coordinates of the preimage multiplied by the scale factor.

1. Triangle NRS is plotted on the coordinate plane.



a. Describe how the image of the dilation of ΔNRS, $\Delta N'R'S'$, compares to the preimage, ΔNRS, using a scale factor of $\frac{2}{3}$ centered at the origin.
The image would be smaller than the preimage and closer to the origin.

b. Complete the table.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point N	$(-9, -3)$	Point N'	$(-6, -2)$
Point R	$(3, -6)$	Point R'	$(2, -4)$
Point S	$(-6, -9)$	Point S'	$(-4, -6)$

c. Draw and label $\Delta N'R'S'$.

d. Use patty paper to verify corresponding angles on the preimage and image have the same measure.

e. Complete the statements.

Triangle NRS and triangle $N'R'S'$ are ○ congruent ○ similar triangles. The side lengths of ○ ΔNRS ○ $\Delta N'R'S$ are $\frac{2}{3}$ times the side lengths of ○ ΔNRS. ○ $\Delta N'R'S'$. The corresponding angles of the triangles are ○ congruent. ○ similar.

9.8.3 BRING IT TOGETHER: DILATIONS ON THE COORDINATE PLANE

DIFFERENTIATE INSTRUCTION

Provide a kinesthetic entry-point by prompting students to sketch the image on patty paper and then align the corresponding angles one at a time to the preimage to allow students to see that the angles fit perfectly one on top of the other.

ENRICHMENT

- How does a scale factor impact the size of the image?
- When does the image get larger? When does it remain the same size? When does it get smaller?

TEACHER MOVES

Remind students that all of the angles are congruent in similar figures and the sides are proportional.

9.8.4 GUIDED PRACTICE: DESCRIBE THE RELATIONSHIP

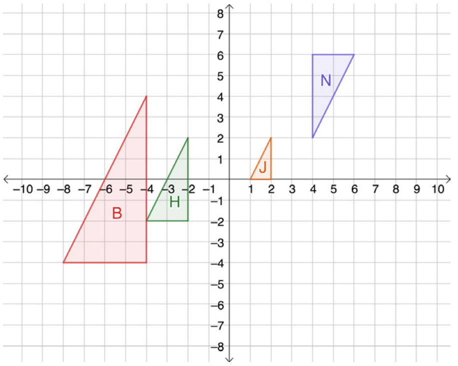
ENRICHMENT

- What makes two figures similar?
- What makes two figures congruent?
- Can more than two figures be similar?
- Can more than two figures be congruent?
- Are similar figures always congruent?
- Are congruent figures always similar?

9.8.4 Guided Practice: Describe the Relationship

Work with your partner to complete the following.

1. Triangles B, H, J, and N are figures on the coordinate plane.



a. Discuss with your partner which triangles are similar or congruent. Then, complete the table.

Triangle Pair	Are they similar?	Are they congruent?
J and N	○ Yes ○ No	○ Yes ○ No
B and H	○ Yes ○ No	○ Yes ○ No
H and N	○ Yes ○ No	○ Yes ○ No
J and B	○ Yes ○ No	○ Yes ○ No

b. Complete the statement.

Triangle *H* is the image of triangle *B* after a dilation using a scale factor of $\frac{1}{2}$ centered at the origin.

c. Explain how you determined your answer to Part B.
I looked at the coordinates of the different triangles, and this was the only pair whose coordinates were proportional.

d. Work with your partner to describe at least three characteristics of the relationship between the triangles selected in Part B.
Answers will vary.

-They are similar.
-Their corresponding angles are congruent.
-Their corresponding sides are proportional.
-Their corresponding sides are proportional by a - scale factor of ½.
-The side lengths of triangle B are 2 times the length of triangle H.
-The side lengths of triangle H are ½ the length of triangle B.

9.8.4 GUIDED PRACTICE: DESCRIBE THE RELATIONSHIP (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students can use patty paper to compare the angles and side lengths.

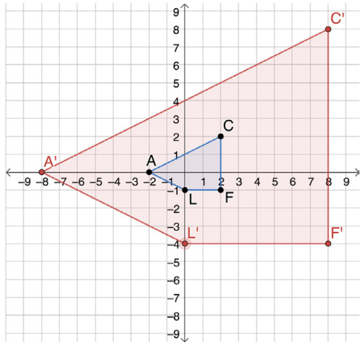
9.8.5 YOUR TURN!

TEACHER MOVES

As you circulate listen for students who are struggling with the connections made during the lesson. Consider working with these students in a small group.

9.8.5 Your Turn!

1. Quadrilateral $ACFL$ is plotted on the coordinate grid.



a. If the figure were dilated by a scale factor of 4 with the origin as the center of dilation, what are the coordinates of the vertices $A'C'F'L'$?

Vertex	Coordinates
A'	$(-8, 0)$
C'	$(8, 8)$
F'	$(8, -4)$
L'	$(0, -4)$

b. Dilate quadrilateral $ACFL$ using a scale factor of 4 centered at the origin. Label the image $A'C'F'L'$.

c. Explain whether the figures are congruent, similar, or neither.
The figures are similar because the side lengths are proportional.

9.8.6 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. Today I am still unsure about …
   Answers will vary.

2. I will try to build my confidence by …
   Answers will vary.

9.8.6 WRAP-UP: GROWTH MINDSET

DIFFERENTIATE INSTRUCTION

Allow students to respond in multiple ways such as in writing, verbally, or by providing visual representations.